Research 2026-10-21: editor, collaboration, diff, offline/PWA, Markdown, import/export (feeds phases 5 and 8)

Versions/dates from the npm registry fetched 2026-10-21 [N]; sizes from Bundlephobia whole-entry min+gzip [B] (approximate, overstate tree-shaken use). GitHub issue counts mostly UNVERIFIED (API blocked). "mine" = researcher's reasoning.

Editors

Editor	Version (last release)	License	Gzip	Markdown round-trip	Health
CodeMirror 6 (view 6.43.15, merge 6.12.3, lang-markdown 6.5.2)	2026-10-19	MIT	view ~80 KB	lossless (the text is the document)	very active; `@uiw/react-codemirror` 4.25.12 supports react >=17
Tiptap 3 (core/react 3.32.0)	2026-10-13	MIT	react ~9, starter-kit ~106, markdown ~19 KB	`@tiptap/markdown` (MarkedJS) is an "early release", no comments, table cells one child node [tiptap.dev/docs/editor/markdown]	very active
Lexical 0.52.0	2026-09-28	MIT	core ~63 KB	transformer based, no loss statement	pre-1.0
BlockNote 0.56.0	2026-10-14	MPL-2.0	~188 KB	docs call Markdown import "lossy"	active
Milkdown 7.22.3 / Plate (platejs 53.3.18) / MDXEditor 4.3.3	Oct 2026	MIT	not measured	fidelity UNVERIFIED	active
Novel 1.0.2	2025-01-18	Apache-2.0	–	Tiptap based	looks stale
Recommendation: source-faithful Markdown with live preview = **CodeMirror 6** (no round-trip risk; AI diffs, 3-way merges and revisions all work on Markdown text; `@codemirror/merge` gives per-hunk accept/reject from the same dependency; preview via `react-markdown` 10.1.0; inline live preview is later polish). Trade-off: not WYSIWYG. Block editing = **Tiptap 3 + @tiptap/markdown, optional, after CM6**, gated: on open, parse then serialize and compare with the stored text; if different, open source mode only. Avoid BlockNote (lossy, biggest, MPL-2.0), Novel (stale), Lexical (pre-1.0). Urdu: CM6 handles bidirectional text (per-line direction from memory: UNVERIFIED, test).					

Collaboration

Yjs 13.6.33 (~28 KB), Hocuspocus 4.7.1 (needs a long-lived server; Vercel WebSocket public beta since 2026-06-22, Hobby availability UNVERIFIED), `@supabase-labs/y-supabase` 0.1.0 (pre-1.0; avoid), Liveblocks (hosted; free plan 500 monthly active rooms, 3 members, non-commercial), Automerge 3.5.0 (~1.6 MB gz: cut). Supabase Realtime without CRDT: 200 concurrent connections, 100 msg/s, 20 presence msg/s, 256 KB broadcast (docs/16) — enough for presence. A CRDT is worth it only if 2+ people routinely type in the same document at once with live

cursors or long offline merges (mine); a Yjs document is binary, so a Markdown projection must be kept in sync for search, llms.txt and AI edits (and the AI would have to edit through Yjs). **Recommendation 5.2:** `version` column, save with `update … where version = base`, on conflict 3-way merge with `node-diff3` 3.2.1 (~2 KB), show remaining conflicts as a diff; Presence only for a soft "X is editing" banner; nothing important in Presence.

Diff and suggestions

`@codemirror/merge` `unifiedMergeView({original})` with `acceptChunk`, `rejectChunk`, `getChunks`; `mergeControls` render function since 6.11.0: best fit for per-hunk accept in the Review inbox (baseline = current doc, editor holds AI text). `diff` (jsdiff) 9.0.0 (~8 KB) for headless hunks (`structuredPatch`, apply selected hunks) and revision compare. diff-match-patch stale (2020): skip. prosemirror-changeset only with ProseMirror: skip. Store suggestions as hunks (base version, range, replacement), not editor marks.

Offline / PWA

`vite-plugin-pwa` 2.0.0 (2026-10-03; breaking changes not read), workbox 7.4.1. Quotas per origin, Chrome up to 60% of disk, Safari 17+ ~60% (MDN storage quotas; webkit.org/blog/14403). Eviction removes an origin's data at once; `navigator.storage.persist()` asks for exemption. Safari deletes script-writable storage after 7 days without interaction; WebKit says home-screen apps have their own counter (webkit.org/blog/10218) but developers report contradictions (bugs.webkit.org 237350): test on your iPhone. Android: assume Chromium rules. Private docs (mine): precache app shell only; do not runtime-cache API responses in Cache Storage; keep an allow-listed set (pinned docs) in IndexedDB via Dexie 4.4.6; offline read-only first (offline editing needs an outbox + merge: a second project). Sign-out: delete IndexedDB databases, `caches.keys()`+`caches.delete()`, unregister the service worker, optionally `Clear-Site-Data: "storage"` (MDN: unregisters service workers; `"cache"` behaviour must be tested). Free plan pause means keep your own JSON backup.

Markdown

CommonMark 0.31.2 (2024-01-28); GFM spec 0.29 (2019), looks stale. Front matter is not standard: `remark-frontmatter` 5.0.0 + `yaml` 2.9.1. unified 11.0.5, remark-parse 11.0.0, remark-gfm 4.0.1, rehype-sanitize 6.0.0, react-markdown 10.1.0: mature, slow-moving. Math: remark-math 6.0.0 + rehype-katex 7.0.1 + katex 0.19.0 (~77 KB, lazy). Mermaid 12.2.0 (~174 KB, lazy-load when a document has a mermaid fence; default `securityLevel: 'strict'`; `sandbox` beta). Sanitization: rehype-sanitize default schema follows GitHub's; place it after the last unsafe step; it prefixes ids/names with `user-content-` (breaks in-page anchors: our "stable block anchors" need a fix-up); for math sanitize first with a schema allowing `math-inline`/`math-display`, then katex; never `allowDangerousHtml` without sanitizing. `[[kind:slug]]` needs a small micromark/remark plugin.

Import / export

`fflate` 0.8.3 (~12 KB) preferred over `jszip` 3.10.2 (~28 KB; CVE-2022-48285 path traversal fixed in 3.8.0). Untrusted zip checklist (SonarSource RSPEC-5042; implementation ours): reject `..`, absolute paths, backslashes, drive letters, NUL; cap entry count, total and per-entry uncompressed size; check compression ratio (legit ≈ 1-3); enforce limits while decompressing; do not trust extensions; run in a Web Worker with a timeout; fflate's built-in limits unknown, add ours. `.docx`: `mammoth` 1.13.0 performs no sanitisation, can emit `javascript:` links, crafted files can burn CPU/memory, Markdown output deprecated: docx -> HTML (mammoth) -> sanitize -> Markdown (`turndown` 7.2.4). Notion export: zip of CSV + Markdown, callouts exported as HTML (notion.com/help/export-your-content); third-party (updated Feb 2026): 32-hex ID suffix per file, images in subfolders, internal links reportedly not preserved (UNVERIFIED): write a resolver. Obsidian: `[[Note]]`, `[[Note|alias]]`,

`[[Note#Heading]]`, `[[Note#^block]]`, `![[embed]]`; on export use standard Markdown links with URL-encoded paths (obsidian.md/help/links).

Stack for phases 5 and 8

1 CodeMirror 6 + @codemirror/merge + react-markdown preview. 2 version-column save + node-diff3 + Presence banner. 3 jsdiff for revision compare/headless hunks. 4 fflate + mammoth + turndown in a Worker with sanitizing and caps. 5 vite-plugin-pwa precaching the shell only + IndexedDB pin list. Cut/defer: CRDT/Hocuspocus/Liveblocks/Automerge, Tiptap block editor (later, gated), offline editing and background sync, inline live-preview decorations, semantic search, Mermaid sandbox, Notion link-graph reconstruction, rich suggestion marks, PDF export (use print CSS). Mistakes to avoid: WYSIWYG that rewrites Markdown on save; caching authenticated API responses in the service worker; not wiping caches/IndexedDB at sign-out; sanitize order mistakes (rehype-raw, KaTeX); unzipping without caps; mammoth Markdown output; relying on y-supabase or Novel; trusting a Safari quota number; assuming a home-screen app is safe from eviction. Not checked: Milkdown/Plate/MDXEditor round-trip; Safari 7-day rule and Clear-Site-Data on real devices; Vercel Hobby WebSockets; jsdiff v9 release notes. Sources: registry.npmjs.org/; bundlephobia.com/api/size; codemirror.net/docs/changelog/ and /docs/ref/#merge; tiptap.dev/docs/editor/markdown; lexical.dev/docs/packages/lexical-markdown; blocknotejs.org/docs/features/import/markdown; github.com/{Milkdown/milkdown, steven-tey/novel, mdx-editor/editor, supabase-community/y-supabase}; liveblocks.io/docs/pricing/plans/free; vercel.com/changelog/websocket-support-is-now-in-public-beta; developer.mozilla.org Storage_quotas_and_eviction_criteria, Clear-Site-Data; webkit.org/blog/{10218,14403}; spec.commonmark.org; mermaid.js.org/config/usage.html; github.com/rehypejs/rehype-sanitize; github.com/mwilliamson/mammoth.js; notion.com/help/export-your-content; obsidian.md/help/links; vulert.com/vuln-db/CVE-2022-48285; rules.sonarsource.com RSPEC-5042.